

DATA ENGINEERING PROJECT - (RAILWAY)

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Internship Project Report

1. Introduction

The railway transport system is an important link for urban areas, an economic driver and a mode of efficient and effective transportation of passengers and goods throughout the country. With so much operational data collected from the railway everyday, data engineering tools are crucial to this process of organizing, cleaning, transforming and analyzing data to get meaningful insights.

This project will include the implementation of data engineering operations on the railway dataset that includes the following data: Train number, Train name, Source station, Destination station, Train operating day. The main purpose of this project is to explore, transform, aggregate, analyze and visualize the data to understand the operational patterns of the railways.

The project starts by loading and examining the data to gain insight into its structure, data types and missing values. A number of data cleaning methods are used to enhance data quality and ensure uniformity of station names and data. The train operations are analyzed based on stations and operating schedules after the train data is preprocessed, filtered and aggregated.

The advanced analytical techniques are also employed to uncover patterns and trends in railway operations. Bar charts, line graphs, and heat maps are used to create visualizations in order to represent the distribution of trains, busiest stations, and operational pattern in an easy to understand format. These visualizations are useful to identify high traffic corridors and train frequencies at various stations.

The project also seeks to generate useful information and suggestions that may be useful for the optimization of railway timetable, station management, and operations. The project showcases the implementation of data engineering principles through the use of Python libraries like Pandas for data manipulation, Matplotlib and Seaborn for visualization, and highlights the analysis of real-world railway data.

2. OBJECTIVE

The project aims to:

1. Gain an understanding of railway operating data.
2. Data cleaning and data preprocessing tasks
3. Look at the trains and what happens at stations
4. Create meaningful visualizations
5. Develop ideas and suggestions

3. DATASET DESCRIPTION

The dataset contains the following columns:

- Train_No
- Train_Name
- Source_Station_Name
- Destination_Station_Name
- days

```
print(df.columns)
```

```
Index(['Train_No', 'Train_Name', 'Source_Station_Name',  
      'Destination_Station_Name', 'days'],  
      dtype='object')
```

4. TOOLS & TECHNOLOGY USED

1. Python:

Throughout the project, data analysis, data transformation and visualization activities were mainly carried out using Python as the main programming language.

2. Pandas:

The railway data data was efficiently loaded, cleaned, filtered, grouped and analyzed using panda library.

3. NumPy:

In the analysis process, numerical operations and handling data-related computations were performed using NumPy.

4. Matplotlib:

Train distribution and station analysis were presented in the form of visualizations using matplotlib like bar charts and line graph.

5. Seaborn:

For advanced data visualization, particularly creating heatmaps and enhancing graphical representation, Seaborn was used.

6. Jupyter Notebook / Google Colab:

Python code was developed, executed and played interactively in the Jupyter Notebook or Google Colab.

5. DATA CLEANING

Data cleaning to enhance the quality of data. The missing data was verified and station names were made consistent (upper case). It is done, as

1. Missing value handling
2. Uppercase conversion
3. Removing spaces

Example:

```
print("checking missing values:\n")
print(df.isnull().sum())
```

checking missing values:

Train_No	0
Train_Name	0
Source_Station_Name	0
Destination_Station_Name	0
days	0
dtype: int64	

```
print("Handle missing values:\n")
df = df.dropna()
```

Handle missing values:

6. DATA EXPLORATION & TRANSFORMATION

Data exploration was used to see which stations were most crowded and what the patterns of train operations were. The analysis of the activity of the source stations was conducted using grouping and aggregation methods.

1. Number of trains
2. Most common stations
3. Saturday filtering
4. Groupby operations

Outputs:

```
print("Displaying the number of trains:\n")
print('Total Trains:', df.shape[0])
```

Displaying the number of trains:

Total Trains: 11113

```
print("Finding Unique Source stations:\n")
print('Unique Source Stations:', df['Source_Station_Name'].nunique())
```

Finding Unique Source stations:

Unique Source Stations: 921

```
print("Finding unique Destination stations:\n")
print('Unique Destination Stations:', df['Destination_Station_Name'].nunique())
```

Finding unique Destination stations:

Unique Destination Stations: 924

```
print("Finding the most common destination Station:\n")
print(df['Destination_Station_Name'].value_counts().head())
```

Finding the most common destination Station:

Destination_Station_Name	
CST-MUMBAI	514
SEALDAH	373
CHENNAI BEACH	342
HOWRAH JN.	337
KALYAN JN	284

Name: count, dtype: int64

```
print("Finding the most common Source Station:\n")
print(df['Source_Station_Name'].value_counts().head())
```

Finding the most common Source Station:

Source_Station_Name	
CST-MUMBAI	513
SEALDAH	372
CHENNAI BEACH	339
HOWRAH JN.	338
KALYAN JN	285

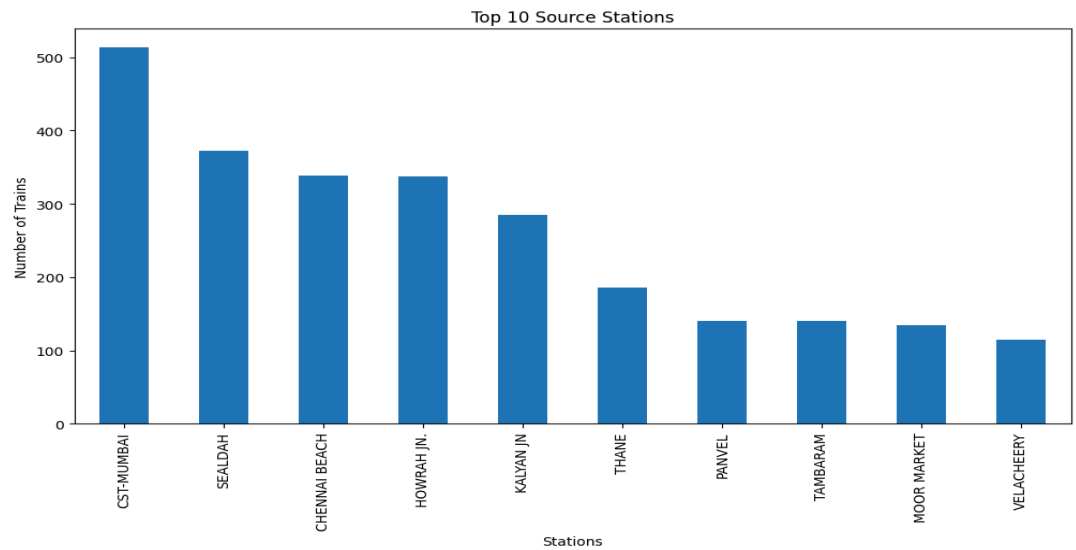
Name: count, dtype: int64

7. VISUALIZATION

The visualization section represents railway operational data using charts and graphs for better understanding and analysis. Multiple visualizations were developed to effectively determine train distribution patterns, busiest stations and operational trends.

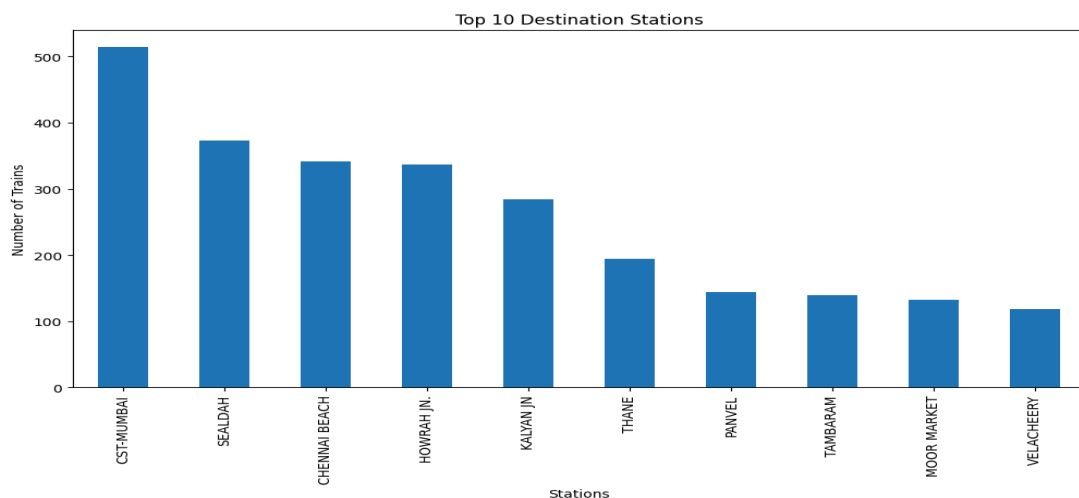
1. Source Station Bar Chart:

This is a bar graph of the stations with the most train departures. It is useful in determining stations with the maximum train operations and passenger movements.



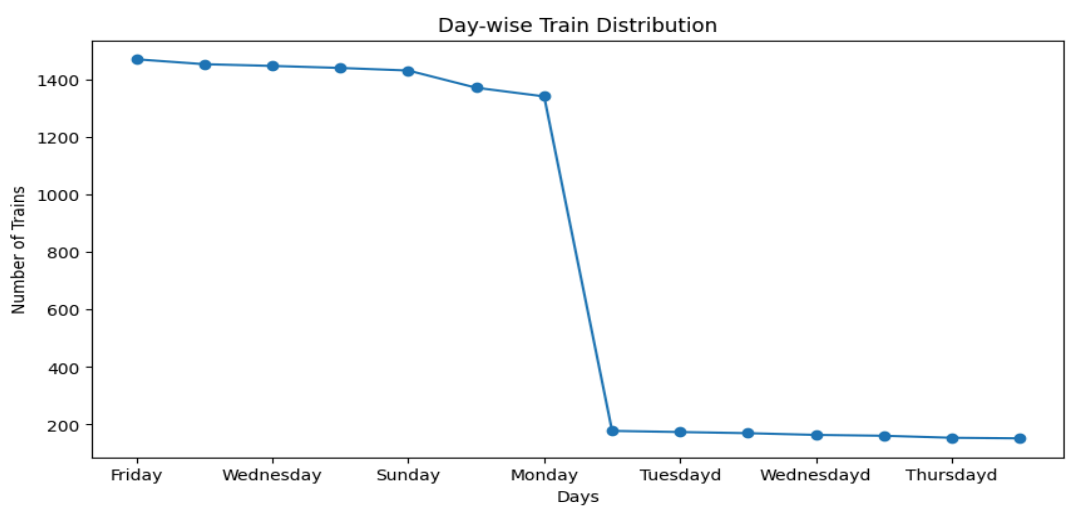
2. Destination station chart:

This bar chart shows the highest ranking stations according to train arrivals. It can be used for analysing stations with a high number of connections and the main railway stations.



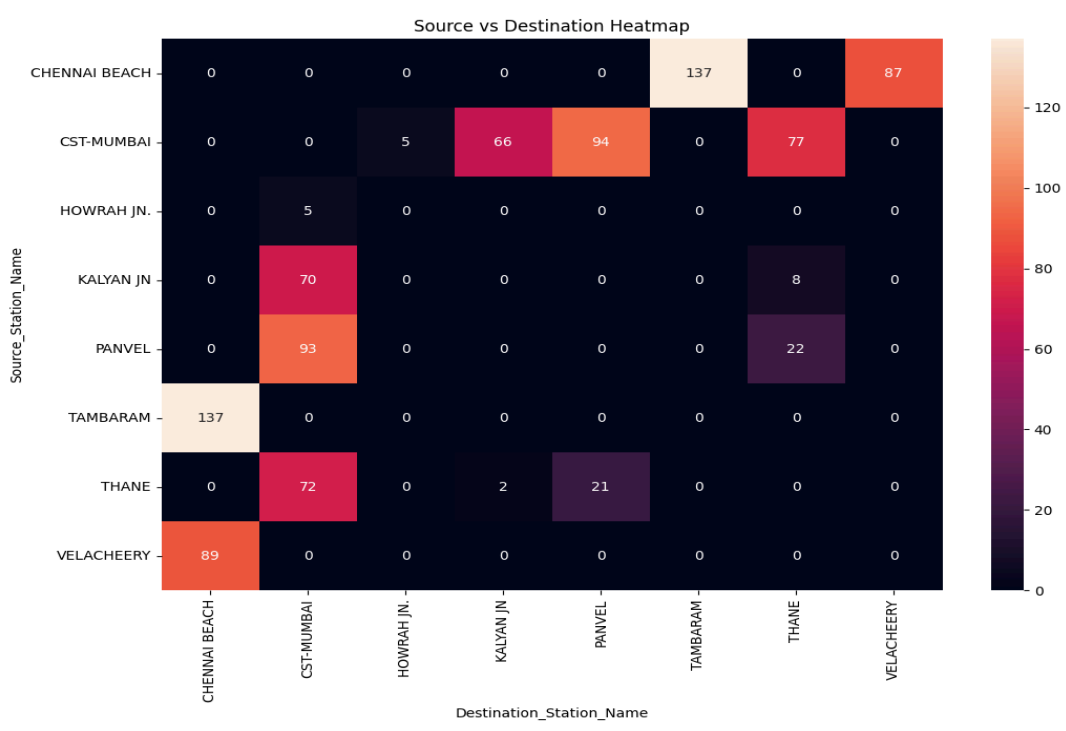
3. Day distribution chart:

The line graph represents the distribution of train operations by day. It helps in understanding operational trends and train activity patterns.



4. Heatmap:

The heatmap is a visualization of the relationship between the source and destination stations. It assists in the identification of very intense railway routes and operational intensity.



8. INSIGHTS & FINDINGS

The railway data was analyzed to identify the key patterns in railway operations and the activity of stations. Some of the stations used as sources and destinations had a large number of train operations, suggesting great importance in the railway network.

Train distribution was visualised and it was found that the distribution of trains differs across the schedules and the routes. Routes were quite busy and some were used more often than others. The analysis also highlighted the importance of proper scheduling and resource management for busy railway stations.

9. RECOMMENDATIONS

1. During peak hours, run more frequent trains on busy lines.
2. Make weekend scheduling more efficient.
3. Improve station operations at busy stations.

10. CONCLUSION

This project has successfully used data engineering techniques for analysing railway operational data. Data was processed through data cleaning, data filtering, data transformation, data aggregation and data visualization to understand the train operation patterns and station activities effectively.

The project gave useful information about the active railway stations, train distribution and active route activity by graphical analysis. In conclusion, the analysis showed that data engineering and visualization methods have the potential to enhance the understanding and decision-making process in railway management systems.
